

(Hyperonic) Fun with ENF

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Abstract

This document explores a few aspects of Holman’s Elegant Normal Form, motivated by applications in the Hyperon context, going beyond the straightforward use of the original ENF for Boolean expressions as developed by Holman. First the theory and practice of applying Holman’s Elegant Normal Form (ENF) to both MeTTa calculus and MeTTa-IL are overviewed, demonstrating how guarded rewrite rules yield unique, minimal process and term representations that preserve semantics while enabling efficient enumeration and search. Next a series of syntax-semantics correlation theorems are developed: from Lipschitz-like bounds for Boolean ENF to average-case linear relations, extensions to general operator alphabets, and preservation (and even improvement) of correlation under Correlated ENF (CENF). A sister paper introduces Semantic ENF (SENF) and its fuzzy, dual-parse variant FUSE-NF—merging CCG and predicate-logic trees with statistical consolidation, background PLN inference, and causal-aware graph transformers. We show here that SENF and FUSE-NF also obey syntax-semantics correlation theorems.

LLMs provided some help here, along with copious human effort

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1 Introduction

Efficient and faithful normalization of symbolic structures is an important aspect of both automated reasoning and advanced AGI architectures. Holman’s Elegant Normal Form (ENF) for Boolean constraint trees demonstrated that a small set of confluent, terminating rewrite rules can turn any Boolean formula into a unique, minimal DAG that preserves its truth-function while enabling near backtracking-free search. This is useful for optimizing SAT search and also within OpenCog’s MOSES probabilistic evolutionary program learning approach.

In this document we extend ENF in two major directions. First, we apply ENF to the MeTTa calculus and to MeTTa-IL, showing how guarded rewrite constructs such as FOR, PAR, NEW, COMM and EQN yield canonical “constraint trees” whose ENF reduction collapses all redundant or impossible branches into a minimal set of guarded actions or equations. We give non-trivial examples, prove termination and confluence, and analyze time and space complexity.

Second, we formally investigate the statistical alignment of syntactic edit distance with semantic distance, which was previously demonstrated only empirically (and only in the Boolean case) in the original work on MOSES. Beginning with Boolean ENF, we prove Lipschitz-like worst-case bounds and, more importantly, average-case linear correlation theorems under mild probabilistic assumptions. We generalize these results to arbitrary discrete operator alphabets (“Extended ENF”) and also connect the results to two sister papers:

- **Correlational ENF (CENF)**, which adapts ENF to specific populations of expressions, optimizing ENF efficiency by clustering strongly co-firing guards into meta-edits
- **Semantic ENF (SENF)**, which adapts ENF to normalize semantic representations of natural language sentences (in a basic form and then a fuzzy form FUSE-NF)

We show that both SENF/FUSE-NF and CENF also encourage syntax-semantics correlation, and that in fact CENF often preserves and often improves the syntax-semantics correlation.

Together, these contributions make progress toward a unified theory and practice of Elegant Normal Form for both logic and language, offering a robust foundation for scalable, semantically grounded AGI systems.

2 ENF for Metta Calculus and Metta-IL

Here we apply the ENF methodology to two formalized language constructs associated with Hyperon.

2.1 ENF for Metta Calculus

2.1.1 Constraint Tree for MeTTa

Given a MeTTa process P , we build a rooted tree $T(P)$ whose internal nodes are the core MeTTa constructs:

PAR, FOR($t < -x$), NEW x , REPL, COMM(t, u), EVAL,

and whose leaves are 0. Each FOR($t < -x$) carries a guard ($t < -x$), and each COMM may later record a unifier σ .

2.1.2 Guard Propagation

Starting from $T(P)$:

1. Push every FOR($t < -x$) node down past PAR and NEW.
2. At each leaf 0, record the conjunction of all guards on the path.

Thus each leaf becomes

$$G_1 \wedge \cdots \wedge G_m : 0,$$

with $G_i = (t_i < -x_i)$.

2.1.3 Local Rewrite Rules

We apply the following confluent, terminating rewrites until no rule applies. Each step removes a node or strictly reduces the size or number of guards.

CommHandle FOR($t < -x$) P | FOR($u < -x$) Q $\rightarrow$ $P\sigma$ | $Q\sigma$,
where $\sigma = \text{unify}(t, u)$.

Replicate REPL{FOR($t < -x$) P } | FOR($u < -x$) Q $\rightarrow$ REPL{FOR($t < -x$) P } | $P\sigma$ | $Q\sigma$.

NewCollapse NEW x in {FOR($t < -x$) P | ...} $\rightarrow$ FOR($t < -x$) P' | ...,
provided $x \notin \text{FV}(\dots)$, with alpha-renaming.

MergeFOR FOR($t < -x$) FOR($u < -x$) P $\rightarrow$ FOR($t \wedge u < -x$) P .

RedundantFOR If P already contains FOR($t < -x$), then FOR($t < -x$) P $\rightarrow$ P .

EvalHandle EVAL(FOR($t < -x$) P) $\rightarrow$ P .

PAR-Elim PAR{ P , 0} $\rightarrow$ P , PAR{0, P } $\rightarrow$ P .

InconGuard If two guards on the same leaf fail to unify, drop that leaf.

2.1.4 MeTTa-ENF Normal Form

Theorem 1. *Every MeTTa process P reduces in a finite number of steps to a unique normal form P_{ENF} . This normal form is confluent, terminating, and semantics-preserving.*

Proof Sketch. Define a weight $W(T) = \sum_{v \in T} (1 + |G(v)|)$. Each rewrite either deletes a node or reduces $|G|$, so termination follows. Local confluence is checked by examining all critical pairs; Newman's lemma then yields global confluence and uniqueness. Soundness holds since each rule is behavior-preserving. $\square$

2.2 Example: ENF on a MeTTa Calculus Program

2.2.1 Original Program

```
NEW x in {
  REPL {
    FOR(pair(a,b)<->x) PAR {
      FOR(head(a)<->h)    COMM<print(h),h>
      FOR(tail(b)<->t)    COMM<process(t),t>
    }
  }
  FOR(pair(a,b)<->x) COMM<logPair(a,b),(a,b)>
}
```

2.2.2 ENFization Steps

1. **Guard Propagation:** Push each FOR guard down to the COMM leaves.
2. **Replication:** Apply the REPL rule to clone the guarded PAR block.
3. **Merge Guards:** Merge identical $\text{FOR}(\text{pair}(a,b) \text{ j-! } x)$ guards.
4. **NEW Collapse:** Eliminate the NEW binding once x no longer scopes over differing guards.
5. **PAR Elimination:** Remove any PAR with a single child.
6. **Redundancy Removal:** Drop any vacuous or duplicate structure.

2.2.3 Resulting ENF Form

```
{pair(a,b)<->x} : COMM<print(h),h>    where h = head(a)
{pair(a,b)<->x} : COMM<process(t),t>  where t = tail(b)
{pair(a,b)<->x} : COMM<logPair(a,b),(a,b)>
```

Here each line is a Start?Stop path in the ENF DAG. The single guard $\{\text{pair}(a,b) \text{ j-! } x\}$ applies to all three actions, and any intermediate NEW, REPL, PAR, or nested FOR nodes have been collapsed away, yielding a unique minimal form that preserves exactly the same reaction semantics.

2.3 More Complex MeTTa Calculus Example

2.3.1 Original Program

```
NEW x in {  
  REPL {  
    PAR {  
      FOR(event(a,b)<->x)  
        PAR {  
          FOR(step1(a)<->s1) COMM<action1(s1),s1>  
          FOR(step2(b)<->s2) COMM<action2(s2),s2>  
        }  
      EVAL(FOR(log(x)<-1) COMM<storeLog(1),1>)  
    }  
  }  
  FOR(event(a,b)<->x) COMM<finalize(a,b),(a,b)>  
}
```

2.3.2 ENF Rewrite Steps

1. **Guard Propagation:** Push each FOR guard down through PAR and NEW so every COMM leaf carries its conjunction of guards.
2. **Replicate:** Apply REPL to clone the guarded block under `event(a,b)<->x`.
3. **MergeFOR:** Merge the two identical `FOR(event(a,b)<->x)` into one.
4. **NewCollapse:** Eliminate the `NEW x in{}` once `$x$` no longer scopes any differences.
5. **PAR-Elim:** Remove any PAR or EVAL wrapper that now has a single child.
6. **Redundancy Removal:** Drop any vacuous or duplicate nodes.

2.3.3 Final ENF Form

```
{event(a,b)<->x} : COMM<action1(s1),s1>   where s1 = step1(a)  
{event(a,b)<->x} : COMM<action2(s2),s2>   where s2 = step2(b)  
{event(a,b)<->x} : COMM<storeLog(1),1>     where 1  = log(x)  
{event(a,b)<->x} : COMM<finalize(a,b),(a,b)>
```

Each line is a Start?Stop path in the ENF DAG. All NEW, REPL, PAR, FOR and EVAL nodes have been collapsed, leaving a single guard and the essential actions.

2.4 ENF on Metta-IL

2.4.1 Constraint Tree Construction

Given a MeTTa-IL compilation unit, parse its AST of:

- **Def** nodes for sort and symbol declarations,
- **Ctor** nodes for term and rewrite constructors,
- **Eqn** nodes for equations $s(\rho) = T \Rightarrow U$,
- **Case** nodes for interaction contexts (application, communication).

Convert this into a rooted *constraint tree* whose internal nodes carry these labels and whose leaves are **Leaf**.

2.4.2 Guard Propagation

Push down type and pattern contexts so that each leaf accumulates a *guard set* of constraints, for example:

$\{\text{sort}(x)=\text{Nat}, \text{ctor}(f)=\text{arity2}, \dots\} : \text{Leaf}.$

2.4.3 Local Rewrite Rules

Apply the following confluent, terminating rewrites until saturation. Each step removes a node or strictly reduces the total weight, where weight is node-count plus sum of guard-set sizes.

InconDef If two **Def** nodes impose inconsistent sort constraints on the same symbol, prune that branch.

PromoteDef Merge nested **Def** nodes by pushing them upward and combining identical declarations.

RedundantCtor Remove a **Ctor** node if its guard set already enforces the same constructor at that position.

OrCaseCut At a **Case** node, split or prune children so that remaining branches have disjoint pattern guards, merging or subsuming overlapping patterns.

AndCaseCut At an **Eqn** node with multiple equations, drop any equation whose guard is implied by another.

ZeroSubsume/OneSubsume If a branch's guard set is empty (always true) or trivially satisfied, inline or remove deeper redundant guards.

MergeEqn Merge sibling **Eqn** nodes whose source/target pairs differ only by alpha-renaming or permutation of non-interacting symbols.

2.4.4 Normal Form and Properties

Theorem 2. *Every MeTTa-IL constraint tree reduces in finitely many steps to a unique normal form, called MeTTa-IL-ENF. The rewrite system is terminating (each rewrite lowers a well-founded weight) and locally confluent; by Newman's Lemma the normal form is unique and semantics-preserving.*

Proof Sketch. Define weight $W(T) = \sum_{v \in T} (1 + |G(v)|)$, where $G(v)$ is the guard set at v . Each rewrite deletes a node or reduces $\sum |G|$, so no infinite sequence exists. Local confluence is verified by examining overlaps of each pair of rules. Since all rewrites preserve the original MeTTa-IL semantics, the final ENF is unique and equivalent. $\square$

2.5 Example: ENF on MeTTa-IL

2.5.1 Original MeTTa-IL Definitions

```
Def sort Bool
Def ctor true   : Bool
Def ctor false  : Bool
Def ctor and    : Bool, Bool -> Bool

Def eqn and(true, true)   => true
Def eqn and(false, _)    => false
Def eqn and(_, false)    => false
Def eqn and(_, _)        => error
```

2.5.2 Constraint Tree Before ENF

$$\text{Eqn}[and] \{ (true, true) \Rightarrow true, (false, _) \Rightarrow false, \\ (_, false) \Rightarrow false, (_, _) \Rightarrow error \}$$

2.5.3 ENF Rewrite Steps

1. *OrCaseCut*: Merge the two branches $and(false, _)$ and $and(_, false)$ into one branch with guard $arg1=false \text{ OR } arg2=false$.
2. *OrCaseCut*: Drop the catch-all branch $(_, _) \Rightarrow error$ since its guard is subsumed by the merged false branch.
3. *Rewrite Saturation*: No further ENF rules apply (guards are now disjoint and minimal).

2.5.4 Final ENF Form

$\{arg1=true, arg2=true\} : true$, $\{arg1=false \text{ OR } arg2=false\} : false$

This gives the unique, minimal ENF for the original MeTTa-IL definitions, preserving exactly the same rewrite semantics.

2.6 More Complex Example of ENF on MeTTa-IL

2.6.1 Original Definitions

```
Def sort Bool
Def ctor true   : Bool
Def ctor false  : Bool
Def ctor not    : Bool->Bool
Def ctor and    : Bool,Bool->Bool
Def ctor or     : Bool,Bool->Bool

Def eqn not(true)   => false
Def eqn not(false)  => true

Def eqn and(true,y) => y
Def eqn and(false,y) => false
Def eqn and(y,z)    => and(y,z)

Def eqn or(true,y)  => true
Def eqn or(false,y) => y
Def eqn or(y,z)     => or(y,z)
```

2.6.2 Constraint Tree Before ENF

$$\begin{aligned} & \text{Eqn}[\text{not}]\{(true) \Rightarrow false, (false) \Rightarrow true\} \\ & \text{Eqn}[\text{and}]\{(true, y) \Rightarrow y, (false, y) \Rightarrow false, (y, z) \Rightarrow \text{and}(y, z)\} \\ & \text{Eqn}[\text{or}]\{(true, y) \Rightarrow true, (false, y) \Rightarrow y, (y, z) \Rightarrow \text{or}(y, z)\} \end{aligned}$$

2.6.3 ENF Rewrite Steps

1. **OrCaseCut on and:** Drop the catch-all branch $(y, z) \Rightarrow \text{and}(y, z)$ since its guard is implied by the first two (true or false cover all Boolean values).
2. **OrCaseCut on or:** Similarly drop the $(y, z) \Rightarrow \text{or}(y, z)$ branch.
3. **Termination:** All remaining guards are disjoint and minimal; no further rules apply.

2.6.4 Final ENF Form

$$\begin{aligned} & \{ \text{not}/1, \text{arg1}=\text{true} \} : false, \\ & \{ \text{not}/1, \text{arg1}=\text{false} \} : true; \\ & \{ \text{and}/2, \text{arg1}=\text{true} \} : \text{arg2}, \\ & \{ \text{and}/2, \text{arg1}=\text{false} \} : false; \\ & \{ \text{or}/2, \text{arg1}=\text{true} \} : true, \\ & \{ \text{or}/2, \text{arg1}=\text{false} \} : \text{arg2}. \end{aligned}$$

Each pair of guards is now disjoint and no redundant catch-all branch remains, yielding the unique minimal ENF for these MeTTa-IL definitions.

3 Syntax-Semantics Correlation Theorems

We now formally demonstrate, for the first time, results corresponding to and then generalizing the empirical observations made in Moshe Looks' PhD thesis, indicating that converting expressions to ENF creates a situation where syntactic distance between expressions correlates with semantic (behavioral) distance.

3.1 Syntax-Semantics Correlation for ENF

We contrast two ways of measuring distance between Boolean functions $f, g : \{0, 1\}^n \rightarrow \{0, 1\}$:

Definition 1 (Semantic Distance). *The semantic distance is*

$$d_{sem}(f, g) = \Pr_{x \in \{0, 1\}^n} [f(x) \neq g(x)].$$

Definition 2 (ENF Edit Distance). *Let T_f, T_g be the unique ENF constraint trees for f, g . Define*

$d_{ENF}(f, g) =$ *the minimum number of local ENF rewrites needed to transform T_f into T_g .*

Suppose every local rewrite in these trees involves a *guard set* of size between $s_{\min}$ and $s_{\max}$.

Theorem 3 (Lipschitz-like Bound for ENF). *Under the above assumptions, the following holds:*

$$2^{-s_{\max}} d_{ENF}(f, g) \leq d_{sem}(f, g) \leq 2^{-s_{\min}} d_{ENF}(f, g).$$

Proof. Each ENF rewrite e_i changes the function value exactly on those inputs in the symmetric difference of two guard sets of size s_i , so its semantic impact is

$$X_i = 2^{-s_i}, \quad s_{\min} \leq s_i \leq s_{\max}.$$

Transforming one tree into the other requires $k = d_{ENF}(f, g)$ such rewrites, each contributing X_i to the overall semantic distance. Since these edits affect disjoint or nearly disjoint input sets,

$$d_{sem}(f, g) = \sum_{i=1}^k X_i.$$

Hence

$$k 2^{-s_{\max}} \leq \sum_{i=1}^k X_i \leq k 2^{-s_{\min}},$$

which gives the stated bounds. $\square$

3.1.1 Why CNF/DNF Fails

In a CNF or DNF representation, a single clause addition or removal can flip the function on up to

$$1 - 2^{-k}$$

of inputs if the clause width is k . Clause sizes can be large, and clauses need not be disjoint. Consequently, no nontrivial Lipschitz-type bound exists: a small syntactic change in CNF/DNF can induce an arbitrarily large semantic change, so syntactic distance in those forms correlates poorly with semantic distance.

3.2 Average-Case Syntax-Semantics Correlation for ENF

Let $f, g : \{0, 1\}^n \rightarrow \{0, 1\}$ have ENF edit-distance

$$k = d_{\text{ENF}}(f, g),$$

and let each local ENF rewrite e_i flip the output on a random set of inputs of measure

$$X_i = \Pr_x[e_i \text{ changes } f(x)], \quad i = 1, \dots, k.$$

Assume:

Assumption 1 (Identical Law and Bounded Support). *The X_i are i.i.d. with*

$$0 \leq X_i \leq 1, \quad \mathbb{E}[X_i] = \mu_0.$$

Assumption 2 (Subgaussian Tail). *There is a constant $c > 0$ such that for all $t > 0$,*

$$\Pr[|X_i - \mu_0| > t] \leq 2 \exp(-c t^2).$$

Define the semantic distance

$$d_{\text{sem}}(f, g) = \Pr_x[f(x) \neq g(x)] = \sum_{i=1}^k X_i,$$

assuming edits affect (approximately) disjoint input subsets.

Theorem 4 (Average-Case Linear Relation). *Under the above assumptions, for any $0 < \varepsilon < 1$,*

$$\Pr\left[|d_{\text{sem}}(f, g) - \mu_0 k| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k).$$

Hence, with probability at least $1 - 2e^{-2\varepsilon^2 k}$,

$$d_{\text{sem}}(f, g) = (\mu_0 \pm \varepsilon) d_{\text{ENF}}(f, g).$$

Proof Sketch. Since the edits are (approximately) independent and each $X_i \in [0, 1]$, Hoeffding's inequality gives

$$\Pr\left[\left|\sum_{i=1}^k (X_i - \mu_0)\right| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k).$$

But $\sum_{i=1}^k X_i = d_{\text{sem}}(f, g)$ and $\sum_{i=1}^k \mu_0 = \mu_0 k$, yielding the result. $\square$

3.3 Average-Case Syntax-Semantics Correlation for Extended ENF

We consider an *Extended ENF* normalization for functions

$$f, g : \mathcal{X} \longrightarrow \mathcal{Y},$$

where $\mathcal{Y}$ is a finite “operator alphabet” (e.g. program or semantic operators). Let $\mathcal{R}$ be a confluent, terminating rewrite system defining Extended ENF, so that each function f has a unique normal-form tree T_f .

Assumption 3 (Semantic Preservation). *Each local rewrite preserves the semantics of the function, i.e. for any rewrite step $T \rightarrow T'$, the underlying functions coincide:*

$$f_T(x) = f_{T'}(x) \quad \forall x \in \mathcal{X}.$$

Assumption 4 (Local Edit Impact). *Each rewrite e_i changes the normal form on exactly those inputs in a set $C_i \subseteq \mathcal{X}$. Under an input distribution μ on $\mathcal{X}$, define*

$$X_i = \mu(C_i) = \Pr_{x \sim \mu}[e_i \text{ flips the output at } x].$$

Assumption 5 (Identical Law and Bounded Support). *The impact variables X_i are i.i.d. with*

$$0 \leq X_i \leq 1, \quad \mathbb{E}[X_i] = \mu_0 > 0.$$

Assumption 6 (Subgaussian Tail and Weak Dependence). *There is a constant $c > 0$ such that for all $t > 0$,*

$$\Pr[|X_i - \mu_0| > t] \leq 2 \exp(-c t^2),$$

and the X_i satisfy the dependency conditions of Hoeffding’s or McDiarmid’s inequality (e.g. bounded dependency graph or block-independence).

Definition 3 (Extended ENF Edit Distance). *Let*

$$d_{\text{ENF}}(f, g) = k$$

be the minimum number of local rewrite steps in $\mathcal{R}$ needed to transform T_f into T_g .

Definition 4 (Semantic Distance). *Define*

$$d_{\text{sem}}(f, g) = \Pr_{x \sim \mu}[f(x) \neq g(x)].$$

Theorem 5 (Average-Case Linear Correlation). *Under the above assumptions, for any $0 < \varepsilon < 1$,*

$$\Pr\left[|d_{\text{sem}}(f, g) - \mu_0 k| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k).$$

Hence with probability at least $1 - 2e^{-2\varepsilon^2 k}$,

$$d_{\text{sem}}(f, g) = (\mu_0 \pm \varepsilon) d_{\text{ENF}}(f, g).$$

Proof Sketch. Confluence and termination guarantee a unique minimal sequence of k local rewrites from T_f to T_g . Each rewrite e_i contributes X_i to the probability of semantic difference, and under weak dependence

$$d_{\text{sem}}(f, g) = \sum_{i=1}^k X_i.$$

Since $X_i \in [0, 1]$ i.i.d. with mean μ_0 , Hoeffding's inequality yields

$$\Pr\left[\left|\sum_i (X_i - \mu_0)\right| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k),$$

giving the result. $\square$

3.4 Preservation of Average-Case Correlation under CENF

Let $f, g : \{0, 1\}^n \rightarrow \{0, 1\}$ have ENF edit distance

$$k = d_{\text{ENF}}(f, g),$$

and suppose we group the k ENF edits into m clusters according to CENF, yielding a CENF edit distance

$$m = d_{\text{CENF}}(f, g).$$

Label the original ENF edits by $e_1, \dots, e_k$ and the clusters by $I_1, \dots, I_m$, where $I_j \subseteq \{1, \dots, k\}$ and $\bigcup_j I_j = \{1, \dots, k\}$.

Assumption 7 (Per-Edit Semantic Impact). *Each original ENF edit e_i flips the function on a random fraction*

$$X_i = \Pr_x[e_i \text{ changes } f(x)],$$

with the X_i i.i.d., $0 \leq X_i \leq 1$, and $\mathbb{E}[X_i] = \mu_0 > 0$.

Assumption 8 (Clustered-Edit Impact). *Define the meta-edit E_j for cluster I_j to affect the union of flips of its members, so*

$$Y_j = \Pr_x\left[x \in \bigcup_{i \in I_j} C_i\right] = \Pr_x\left[\bigvee_{i \in I_j} (e_i \text{ flips at } x)\right].$$

Assumption 9 (Subgaussian Tails and Weak Dependence). *There is $c > 0$ such that each X_i satisfies*

$$\Pr[|X_i - \mu_0| > t] \leq 2 \exp(-ct^2),$$

and the X_i (and hence the Y_j) obey a bounded-dependency condition allowing Hoeffding-type concentration.

Theorem 6. *Under these assumptions, the semantic distance*

$$d_{\text{sem}}(f, g) = \Pr_x[f(x) \neq g(x)] = \sum_{j=1}^m Y_j$$

satisfies, for any $0 < \varepsilon < 1$,

$$\Pr\left[|d_{\text{sem}}(f, g) - \mu_0 k| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k).$$

Hence with probability at least $1 - 2e^{-2\varepsilon^2 k}$,

$$d_{\text{sem}}(f, g) = (\mu_0 \pm \varepsilon) d_{\text{ENF}}(f, g) = (\mu_0 \pm \varepsilon) \sum_{j=1}^m |I_j|.$$

Proof Sketch. By construction, each meta-edit E_j encompasses the union of its $|I_j|$ original flips, so in expectation

$$\mathbb{E}[Y_j] = \mathbb{E}\left[\sum_{i \in I_j} X_i\right] = |I_j| \mu_0.$$

Summing over $j = 1, \dots, m$ gives

$$\mathbb{E}\left[\sum_{j=1}^m Y_j\right] = \mu_0 \sum_{j=1}^m |I_j| = \mu_0 k.$$

Each Y_j is bounded in $[0, 1]$ and (under weak dependence) Hoeffding's inequality yields

$$\Pr\left[\left|\sum_j (Y_j - \mathbb{E}[Y_j])\right| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k).$$

Since $d_{\text{sem}}(f, g) = \sum_j Y_j$, the result follows. $\square$

3.5 When CENF Improves Average-Case Syntax-Semantics Correlation

Although CENF is designed to preserve the tight linear relation between edit-distance and semantic-distance established by ENF, it can in fact *enhance* that correlation in practice. The following situations illustrate why clustering ENF edits into meta-edits often yields a stronger average-case alignment:

- **Variance reduction via clustering.** Raw ENF edits e_i may have highly varying semantic impacts X_i . By grouping correlated edits into meta-edits E_j with impact

$$Y_j = \sum_{i \in I_j} X_i,$$

the variance of each Y_j is lower (by the central limit effect) than that of individual X_i . Summing Y_j yields a total semantic distance that concentrates more tightly around its mean, improving correlation with d_{CENF} .

- **Elimination of spurious micro-edits.** Some ENF operations merely reorder independent branches or merge redundant guards, causing zero or negligible semantic change. CENF collapses these into single meta-nodes or drops them altogether, so d_{CENF} no longer counts trivial edits. Removing such noise raises the signal-to-noise ratio in the edit-distance versus semantic-distance relationship.
- **Alignment with true semantic modules.** When a set of ENF edits always co-fires as a semantic unit—a "schema" or building block—CENF’s clustering recovers that unit as one meta-edit. Measuring meta-distance then directly counts the number of differing modules, yielding near-perfect per-edit semantic weight and hence stronger average correlation.
- **Stabilized Lipschitz slope.** In ENF the worst-case bounds

$$2^{-s_{\max}} d_{\text{ENF}} \leq d_{\text{sem}} \leq 2^{-s_{\min}} d_{\text{ENF}}$$

can be very loose if guard sizes vary widely. CENF typically produces meta-guards of more uniform effective size s' , narrowing the gap between $2^{-s_{\max}}$ and $2^{-s_{\min}}$, and thus tightening the bound and improving average-case behavior.

In these cases, the edit-distance under CENF aligns more faithfully—and with higher statistical consistency—with the true semantic distance than raw ENF, yielding better performance in downstream applications.

3.6 Practical Examples Where CENF Improves Correlation

3.6.1 Grouped Building-Block Schemata

Consider two Boolean functions $f, g : \{0, 1\}^4 \rightarrow \{0, 1\}$ defined by

$$f(x) = (x_1 \wedge x_2) \vee (x_3 \wedge x_4), \quad g(x) = (\neg x_1 \wedge \neg x_2) \vee (x_3 \wedge x_4).$$

Under raw ENF, one must flip both literals in the first conjunct, so

$$d_{\text{ENF}}(f, g) = 2, \quad d_{\text{sem}}(f, g) = \Pr[f(x) \neq g(x)] = \frac{3}{16} \approx 0.1875.$$

Hence each ENF edit carries semantic weight ≈ 0.09375 .

By clustering the pair (x_1, x_2) into a single meta-guard and likewise (x_3, x_4) , CENF uses one meta-edit:

$$d_{\text{CENF}}(f, g) = 1, \quad d_{\text{sem}}(f, g) = 0.1875,$$

so the per-edit semantic weight doubles to 0.1875, improving the alignment.

3.6.2 Pruning Spurious Micro-Edits

Now let

$$h(x) = (x_1 \vee x_2) \vee (x_1 \vee x_2), \quad k(x) = (x_2 \vee x_1) \vee (x_1 \vee x_2).$$

These two formulas are semantically identical, so

$$d_{\text{sem}}(h, k) = 0.$$

Raw ENF nonetheless counts two commutation steps (swapping the first disjunct’s literals):

$$d_{\text{ENF}}(h, k) = 2, \quad \text{per-edit weight} = 0.$$

CENF collapses duplicate branches and recognizes commutativity, yielding

$$d_{\text{CENF}}(h, k) = 0, \quad d_{\text{sem}}(h, k) = 0,$$

perfectly matching syntax-distance to semantic-distance and thus removing spurious edits.

3.7 Average-Case Syntax-Semantics Correlation for SENF

A sister paper introduces Semantic ENF (SENF) and its fuzzy dual-parse variant FUSE-NF, which merges CCG-derived semantic trees with LLM-generated predicate-logic parses into a single cross-linked graph. Statistical pattern-mining (frequent subtree, role-clustering, mutual information, paraphrase alignment, autoencoder analysis) yields data-driven consolidation rules, while background Probabilistic Logic Network inference injects KB entailments and causal-association discovery adds directed edges. The resulting fuzzy-SENF graph is encoded by a causal-aware graph transformer, providing a rich, yet compact, representation for downstream tasks such as SAT-style enumeration, probabilistic evolutionary learning in MOSES, and causal reasoning within the Hyperon AGI framework. Here we show that SENF and FUSE-NF also obey formal syntax-semantics correlation theorems.

SENF and FUSE-NF encourage syntax-semantics correlation just like the other forms of ENF. Let s_1, s_2 be two sentences, and let

$$d_{\text{SENF}}(s_1, s_2) = k$$

be the minimum number of local SENF edits (on the combined CCG and logic tree, cross-links, consolidations, etc.) required to transform the SENF graph of s_1 into that of s_2 . Define the *semantic distance*

$$d_{\text{sem}}(s_1, s_2) = \Pr_{x \sim \mu} [s_1(x) \neq s_2(x)],$$

where μ is a distribution over input contexts.

Assumption 10 (Uniform Local Impact). *Each local SENF edit e_i flips the output on a random set of contexts of measure*

$$X_i = \Pr_{x \sim \mu} [e_i \text{ changes } [s](x)], \quad i = 1, \dots, k,$$

where the X_i are identically distributed, satisfy

$$0 \leq X_i \leq 1, \quad \mathbb{E}[X_i] = \mu_0 > 0.$$

Assumption 11 (Subgaussian Tails and Weak Dependence). *There exists $c > 0$ such that for all $t > 0$,*

$$\Pr[|X_i - \mu_0| > t] \leq 2 \exp(-c t^2),$$

and the joint law of the X_i admits a Hoeffding-type concentration bound (e.g. bounded dependency or block-independence).

Theorem 7 (Average-Case Linear Correlation). *Under these assumptions, for any $0 < \varepsilon < 1$,*

$$\Pr[|d_{\text{sem}}(s_1, s_2) - \mu_0 k| > \varepsilon k] \leq 2 \exp(-2\varepsilon^2 k).$$

Hence with probability at least $1 - 2 \exp(-2\varepsilon^2 k)$,

$$d_{\text{sem}}(s_1, s_2) = (\mu_0 \pm \varepsilon) d_{\text{SENF}}(s_1, s_2).$$

Proof Sketch. By confluence and termination of the SENF rewrite system, there is a unique minimal sequence of k edits transforming one SENF graph into the other. Each edit’s semantic impact is X_i , and under weak dependence

$$d_{\text{sem}}(s_1, s_2) = \sum_{i=1}^k X_i.$$

Since $X_i \in [0, 1]$ i.i.d. with mean μ_0 , Hoeffding’s inequality gives the stated exponential-tail bound, yielding the linear average-case correlation. $\square$

3.8 Average-Case Syntax–Semantics Correlation for FUSE-NF

Let s_1, s_2 be two sentences and let

$$d_{\text{FUSE}}(s_1, s_2) = k$$

be the minimal number of local edits—across the combined CCG branch, logic branch, alignment edges, consolidation rules, PLN injections and causal-association edges—needed to transform the FUSE-NF graph of s_1 into that of s_2 . Define the semantic distance

$$d_{\text{sem}}(s_1, s_2) = \Pr_{x \sim \mu} [[s_1](x) \neq [s_2](x)],$$

where μ is a distribution over input contexts.

Assumption 12 (Uniform Local Impact). *Each local edit e_i in the FUSE-NF graph flips the output on a random fraction of contexts of measure*

$$X_i = \Pr_{x \sim \mu} [e_i \text{ changes } [s](x)], \quad i = 1, \dots, k,$$

and the X_i are identically distributed with

$$0 \leq X_i \leq 1, \quad \mathbb{E}[X_i] = \mu_0 > 0.$$

Assumption 13 (Subgaussian Tails and Weak Dependence). *There exists a constant $c > 0$ such that for all $t > 0$,*

$$\Pr[|X_i - \mu_0| > t] \leq 2 \exp(-c t^2),$$

and the joint distribution of the X_i satisfies the conditions for Hoeffding's (or McDiarmid's) inequality under bounded dependency or block-independence.

Theorem 8 (Average-Case Linear Correlation). *Under these assumptions, for any $0 < \varepsilon < 1$,*

$$\Pr\left[|d_{\text{sem}}(s_1, s_2) - \mu_0 k| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k).$$

Hence with probability at least $1 - 2 \exp(-2\varepsilon^2 k)$,

$$d_{\text{sem}}(s_1, s_2) = (\mu_0 \pm \varepsilon) d_{\text{FUSE}}(s_1, s_2).$$

Proof Sketch. Confluence and termination of the FUSE-NF rewrite system guarantee a unique minimal sequence of k local edits transforming the graph of s_1 into that of s_2 . Each edit contributes a random semantic impact X_i , and under weak dependence

$$d_{\text{sem}}(s_1, s_2) = \sum_{i=1}^k X_i.$$

Since $0 \leq X_i \leq 1$ are i.i.d. with mean μ_0 , Hoeffding's inequality implies

$$\Pr\left[\left|\sum_{i=1}^k (X_i - \mu_0)\right| > \varepsilon k\right] \leq 2 \exp(-2\varepsilon^2 k),$$

yielding the stated bound. □